



# NOVA HOME

Current Conditions → Explainable AI Recommendation → Future Fan-Demand Prediction

## Media #10 · AI Prediction Prototype



**SIMULATION ONLY**

**Demo prediction values — NOT real model output**

*Outputs displayed are prototype mock parameters.*

## Current Conditions

Simulated sensor state • Demonstration values only

 Temperature

28.4 °C

 Humidity

61 %

 Light Level

42 %

 Motion

Detected

 Window

Closed

## AI Recommendation

Explainable decision logic based on current room state

MAIN RECOMMENDATION

 **Turn ON the Fan**

### Why?

Temperature is elevated while the window is closed. The AI model uses recent sensor patterns and environmental conditions to estimate whether fan operation will be needed.

 Recommendation is based on simulated conditions in this prototype.

## Predicted Fan Demand

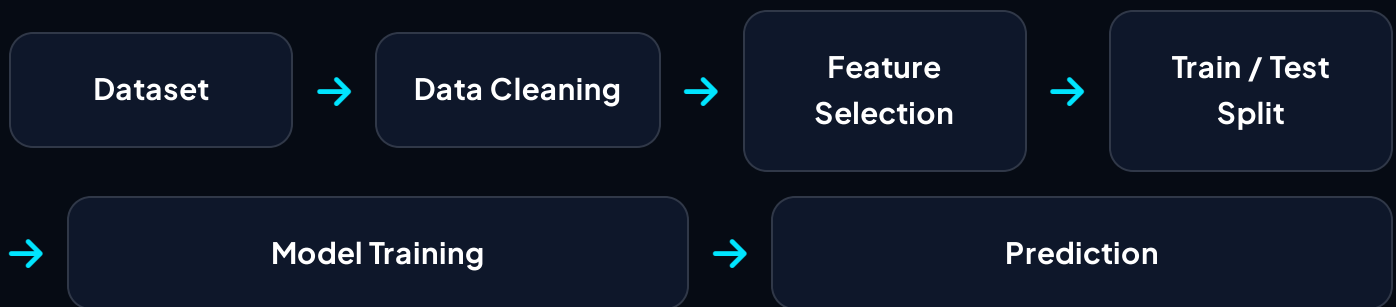
Estimated demand over the next 30 minutes



 Simulated prediction values — demonstration only

## AI / Machine Learning Pipeline

End-to-end data transformation pipeline producing the AI prediction



 Python + scikit-learn

## Model Input Features

Smart-home telemetry feature vector

Temperature

Humidity

Light\_Level

Motion

Door

Window

Fan

Light

User\_Command

Timestamp

*Source: NOVA HOMEESP32 dataset (Values displayed in this prototype are simulated unless a real exported ESP32 dataset is connected).*

## Prediction Logic

The model learns relationships between environmental conditions, device states, and historical fan usage. After training and evaluation, the model can estimate future fan demand from the latest sensor conditions.

 [Example implementation](#)

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**Important Notice:** The values shown in this prototype are simulated/mock data for demonstration. They are not real sensor measurements and must be replaced with outputs from the trained model.